

The Keel and the North Star (Overview)

How to use this narrative

This is the spoken story behind the 14-slide deck, written to be read aloud or adapted. Slides 7, 8, 9, 10 and 12 get full treatment because they carry the most detail and the most acronyms. Every acronym is spelled out the first time it appears in a section, and the glossary at the end collects them all.

Part	Slides	Audience	Suggested time
The metaphor	1–6	Everyone	8 min
For business executives	7–8 (after 4–6)	Executives	7 min
For technical leaders	9–12	Technical leaders	12 min
Roadmap and decisions	13–14	Everyone	5 min

In a mixed room, run the executive section in full and move quickly through slides 9–12, pausing on slide 12 because it answers the integration question most leaders arrive with.

The opening arc: slides 1–6

The first six slides give everyone one picture to hold: a ship with a compass, sails, a keel and a star to steer by.

Slide 1, The Keel and the North Star. Insperity's logo was designed as a compass, so the company already navigates by one. This talk is about what keeps the ship on that heading. Name the two audiences up front: business executives care about value, risk and outcomes; technical leaders care about how it works, how it stays vendor-neutral and how it ships.

Slide 2, the Keel. Walk the picture top to bottom. Compass is the steering instrument at the helm: it sets direction, policy and guardrails. Copilot is the sails: Jarvis (Insperity's AI assistant), domain agents and Microsoft 365 Copilot doing the work. The Keel sits below the waterline: the holonic semantic layer. Executives rarely see it, yet everything above depends on it.

The physics carries the message. A sailboat without a keel is pushed sideways by the wind. With a keel, the same wind drives it forward. For Insperity the wind is regulatory change across 50 states, new AI hiring laws, new platforms and new AI agents. The bulb at the bottom of the keel is the ballast: the L1 (Level 1) knowledge holons, meaning the 142 policy rules, benefit plan provisions and admitted records that give the structure its weight.

Slide 3, the wind is already blowing. Four numbers set the scale: roughly 300,000 worksite employees across thousands of client companies, more than 50 state and local jurisdictions, 142 rules in the policy register across five service domains, and at least four platforms with their own definitions (Workday, Insperity payroll, Microsoft 365 and Box). Each platform and each AI agent defines "eligible employee" or "plan year" its own way. That drift is the sideways force.

Slide 4, three promises. Reduce the Keel to three commitments. One meaning: worker, client, plan, jurisdiction and rule mean the same thing everywhere. Author once, enforce everywhere: a law change is written once and compiled into every system that runs the business, with a target of 48 hours from law change to production. Every answer traceable: each payroll check, AI answer and agent action carries the rule and statute behind it.

Slide 5, the North Star. The compass mark becomes the front door to everything the company knows. Tap it in a portal, in Microsoft Teams or in Jarvis, and you are searching governed knowledge. Every question resolves to five coordinates, the same five the governance gateway enforces: WHY at north (the rule and the law behind it), WHAT (services, processes, plans), WHO (clients, employees, roles, AI agents), WHERE (jurisdictions, worksites, client tenants) and WHEN on the outer ring (effective dates, plan years, rule versions).

Slide 6, from the mark to the answer. Follow one question. A Colorado employee asks how much paid family leave they can take. The Keel resolves the five coordinates, searches only admitted knowledge, filters by the employee's client and role, checks the answer's reliability and cites its sources. The employee sees their entitlement under Colorado FMLI (Family and Medical Leave Insurance). The HR business partner sees eligibility and the payroll rule. The executive sees an aggregate with no personal data. Same truth, shaped by who asks.

Slide 7: What each group gets

Each of the four groups gets a concrete change and a number to hold the program to. All targets are planning estimates, measured against baselines set in Phase 0.

Worksite employees. A worksite employee is a person employed through Insperity's co-employment arrangement: they work for the client company, and Insperity handles payroll, benefits and compliance as co-employer. Today many of their questions go to a service center.

With the Keel, Jarvis answers pay, benefits and leave questions in seconds and cites the plan provision or statute. Training arrives when a rule or their role changes, not once a year. The measure is self-service resolution, the share of questions answered correctly without a person stepping in: about 40% at the end of Phase 1, rising to 80% by Phase 3.

Client employers. These are the owners, HR leads and managers of the small and midsize businesses Insperity serves. They receive a new product: a monthly Compliance Assurance Statement listing the rules applied to their workforce, the law changes absorbed that month, training completed and open items. Behind it, a new state or local law is written once and reaches their payroll within days, then within hours. The measure is time from law change to compiled policy in production: weeks today, 10 days in Phase 1, 5 days in Phase 2 and 48 hours or less in Phase 3.

Service teams. This group includes BPAs (Business Performance Advisors, Insperity's client-facing sales and advisory staff), HR specialists, payroll and benefits specialists, and service representatives. Their prep work, proposal drafts and case summaries are grounded in governed knowledge, so they are already checked against each client's jurisdictions. The measure is a proposal-prep time index that starts at 100 in Phase 0 and targets 50 by Phase 3, meaning half the time.

Insperity enterprise. For executives, compliance and finance, the change is evidence that assembles itself. Three attestations matter most:

- **CPEO** (Certified Professional Employer Organization): an IRS certification under Internal Revenue Code section 7705 that lets a PEO take on federal employment tax liability for clients. It requires quarterly attestations and a surety bond.
- **SOC** (System and Organization Controls) reports: independent audits under AICPA (American Institute of Certified Public Accountants) standards. SOC 1 covers controls relevant to clients' financial reporting, such as payroll. SOC 2 covers security, availability, confidentiality and privacy.
- **ESAC** (Employer Services Assurance Corporation): an independent accreditation body for PEOs that checks financial health, ethics and client protection.

The measures are an audit-evidence time index (100 at baseline, 38 by Phase 3) and zero cross-tenant incidents. A cross-tenant incident is any case where one client company's data reaches another client's users or AI answers. Zero is a hard gate at every phase, not a target.

Slide 8: The cost of sailing without a keel

The Keel is a risk control as much as an accelerator, and this slide names four risks that grow every quarter without it.

Agent sprawl. AI agents are arriving from every direction: Insperity's own HR360 agents, agents built in Microsoft Copilot Studio, and agents inside Workday. Without a keel, each one answers from its own copy of the truth, grounded in whatever documents its builder connected.

With the Keel, every agent is registered, has its own identity in the company directory, and reaches data only through one governance gateway. An unregistered agent simply cannot reach governed knowledge.

Definition drift. Ask Workday, the payroll system, Microsoft Fabric (Microsoft's analytics platform) and Box (the content repository) what an "eligible employee" is, and today you may get four different answers. Reports disagree, and AI answers disagree with the reports. With the Keel, there is one canonical definition, and the versions each platform uses are generated from it rather than maintained by hand.

Audit exposure. Before each CPEO (Certified Professional Employer Organization), SOC (System and Organization Controls) or ESAC (Employer Services Assurance Corporation) review, teams rebuild evidence by hand from many systems. Every new system adds to that burden. With the Keel, evidence is produced continuously as a byproduct of normal operation, and each item is tied to the rule it proves.

Vendor lock-in. If the company's knowledge lives in one vendor's proprietary format, that vendor effectively owns the roadmap. With the Keel, knowledge is kept in open standards, and any vendor adapter can be replaced within a quarter without rewriting the knowledge. The program tests one such swap at every phase gate, so the claim is proven rather than asserted.

Close the executive section here: the Keel answers each risk with a specific control, not a policy memo.

Slide 9: Inside the Keel, five levels

The Keel is built as a holarchy: five levels, each with a defined contract to the levels around it, so any one part can change without breaking the others.

A **holon** is something that is a whole in itself and also part of something larger, the way a department is complete on its own yet part of a company. A **holarchy** is a hierarchy of holons. The word "contract" on this slide means a fixed agreement about what one level provides to its neighbours.

L4, Enterprise holarchy (Level 4). The top level holds what the whole company shares.

- **isp-core** is Insperity's core ontology: the canonical definitions of client company, worker, jurisdiction, employment type and statutory standard. An **ontology** is a formal, machine-readable model of the concepts in a business and how they relate.
- **SKOS** (Simple Knowledge Organization System) is a W3C standard for vocabularies, taxonomies and thesauri, used here for controlled terms and their synonyms. **W3C** is the World Wide Web Consortium, the body that sets web standards.
- **ORSD** (Ontology Requirements Specification Document) templates state what each part of the model must be able to answer, written as competency questions.

- The **Semantic Governance Council** is the cross-functional body (legal, compliance, domain owners, architecture) that approves changes.
- Its contract: domains inherit isp-core through **owl:imports**, the mechanism in OWL (Web Ontology Language) by which one ontology includes another. Every change is recorded in **KGCL** (Knowledge Graph Change Language), a standard format for describing changes to an ontology so they can be reviewed and traced.

Mediation holon. This level sits beside the others and connects them to the outside world.

- The **mapping graph** records how each Insperity concept corresponds to fields in Workday, payroll, analytics tools and search indexes.
- The **identity and URI registry** gives every worker, client and rule one permanent identifier. A **URI** (Uniform Resource Identifier) is a web-style unique name, so the same employee in Workday and in payroll resolves to one identity.
- **Harmonization** resolves conflicts between federal, state and local rules, applying the most protective rule for the employee.
- **RAG anchors** tie AI retrieval to those permanent identifiers. **RAG** (retrieval-augmented generation) is the technique of giving an AI model relevant documents to ground its answer.
- Its contract: six defined correspondence types (for example, exact match or broader-than) and lineage from every generated artifact back to its source.

L3, Gateway (Level 3). This is where Compass runs at the moment a request is made.

- The **5D context bundle** captures the five dimensions of every request: WHO, WHAT, WHEN, WHERE and WHY.
- **Policy enforcement** happens at a PEP (policy enforcement point), the checkpoint that allows, denies, holds or redacts each request.
- The **MTBH circuit breaker** stops unreliable AI answers. **MTBH** (Mean Time Between Hallucinations) is the program's reliability measure, scored across the five dimensions; a hallucination is an AI answer that is confident but wrong. When a score falls below threshold, the breaker trips and routes the case to a person.
- The **Simplex gate** holds any irreversible action, such as changing a payroll record or remitting taxes, until a person approves it. The name comes from the Simplex safety architecture, where a trusted, simple controller oversees a more complex one.
- Its contract: every agent call passes one policy decision point, with no side doors.

L2, Domain holons (Level 2). Each service domain is sovereign, with its own ontology, AI agent, search index and monitors. The domains are isp-pay (payroll and tax), isp-ben (benefits), isp-comp (compliance and risk), isp-tal (talent), isp-tech (HR technology), plus Client and Contractor. Adding a new domain never touches the existing ones.

L1, Knowledge holons (Level 1). The atomic unit: one statute clause, one benefit plan provision or one handbook policy. Each has a URI, passes **SHACL** validation, is versioned with KGCL, and is linked to the legal obligation that governs it. **SHACL** (Shapes Constraint

Language) is a W3C standard for validating that data meets defined rules, such as "every plan provision must have an effective date."

Slide 10: Author once, compile everywhere

A rule is written once in a canonical form, and a compiler generates every version the business systems need, each one pointing back to the statute.

The canonical rule. The slide shows California daily overtime (register rules PAY-005 and PAY-006) as written once:

- `isp-pay:CADailyOvertimeNorm` is the rule's permanent name in the payroll domain.
- `a lkif:Obligation` says the rule is a legal obligation. **LKIF** (Legal Knowledge Interchange Format) is an ontology for legal concepts such as obligations, permissions and prohibitions.
- `lkif:isSubjectTo "CA Labor Code §510"` names the statute it comes from. **CA** is California.
- `isp-core:jurisdiction "US-CA"` sets where it applies.
- `dailyThresholdHours 8.0` and `rateMultiplier 1.5` carry the logic: hours over 8 in a day are paid at 1.5 times the regular rate. The full rule also carries the double-time tier over 12 hours.

This is written in **RDF** (Resource Description Framework), the W3C standard for representing knowledge as linked statements. Alongside it sits a SHACL (Shapes Constraint Language) shape that states how to check compliance.

The five compiled targets. The compiler reads the canonical rule and generates:

1. **dbt / SQL tests** for batch gross-to-net checks. **dbt** (data build tool) is a widely used framework for testing and transforming data in warehouses. **SQL** (Structured Query Language) is the standard database query language. Gross-to-net is the payroll calculation from gross pay to take-home pay. These tests scan millions of timesheet rows per run.
2. **OPA / Rego** policy at the gateway. **OPA** (Open Policy Agent) is an open-source engine that makes allow or deny decisions; **Rego** is its policy language. Decisions take under a millisecond.
3. **JSON Schema** constraints for Jarvis and other AI agents. **JSON** (JavaScript Object Notation) is the common data format for APIs; JSON Schema defines what a valid response must contain, so an agent cannot return an answer missing its rule citation.
4. **Workday configuration**: validations and business-process conditions for clients on HRScale, the Insperity service built on Workday.
5. **Fabric IQ ontology** for analytics and AI grounding. Fabric IQ is the semantic layer in Microsoft Fabric that lets reports and agents share business definitions.

Why it matters. Every generated artifact carries the source shape's URI (Uniform Resource Identifier). A rejected timesheet therefore points straight to its statute, and an auditor can follow the chain in one step. Nobody edits a generated version by hand; a drift check in CI (continuous integration, the automated build-and-test pipeline) flags any difference. Because the runtime systems execute compiled rules rather than reasoning over the knowledge graph, there is no speed penalty.

The timing target that follows from this: a law change reaches production in 10 days in Phase 1 and 48 hours or less by Phase 3.

Slide 11: How North Star search works

Search is the Keel exposed as navigation, in six steps. Nothing can be found until it has been admitted. Each item is classified, tagged with its client tenant and its PII (personally identifiable information) class, and SHACL-validated. Admitted items are anchored to URIs, and each question is resolved into the five coordinates. Retrieval then combines SPARQL graph queries with vector search. The policy enforcement point filters out anything outside the asker's tenant or role. Finally MTBH scores the answer, uncertain answers go to a person, and every answer cites its sources. **SPARQL** (SPARQL Protocol and RDF Query Language) is the standard query language for knowledge graphs. On Microsoft, this runs on Foundry IQ and Azure AI Search behind API Management; the six steps are the same on any stack.

Slide 12: Stack independent by contract

The Keel owns contracts, not products: eight ports, each defined by open standards, with Microsoft as the first adapter set and other vendors as proven alternates.

A **port** is a fixed interface the Keel exposes; an **adapter** is the product that plugs into it. The canonical contract column never changes. Adapters can be swapped, and the program tests one swap at every phase gate. This is the answer to integration and interoperability concerns: Insperity can run on Microsoft today, keep Workday as the system of record for HRScale clients, and change model or cloud providers later without rewriting its knowledge.

Port	What it carries	Canonical contract	Microsoft adapter	Alternates
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Meaning	Entities, vocabularies, definitions	OWL 2 (Web Ontology Language), SKOS (Simple Knowledge Organization System), LKIF (Legal Knowledge Interchange Format)	Fabric IQ ontology, generated from the canonical model	Amazon Neptune (graph database), GraphDB (RDF graph database)
Rules	Obligations, thresholds, constraints	SHACL (Shapes Constraint Language), SWRL (Semantic Web Rule Language), KGCL (Knowledge Graph Change Language)	Fabric and dbt data tests; Power Automate conditions	Workday validations; dbt tests
Policy	Allow, deny, hold, redact decisions	OPA (Open Policy Agent) with Rego; JSON Schema	API Management calling an OPA decision point; Purview DLP (data loss prevention)	Cedar (AWS's open-source policy language); Envoy (open-source proxy)
Identity	People, clients, agents, identifiers	OIDC (OpenID Connect), SCIM (System for Cross-domain Identity Management), URI registry	Entra ID (formerly Azure Active Directory); Entra Agent ID for AI agents	Okta; Keycloak (open source)
Grounding	Admitted knowledge for AI retrieval	SPARQL graph queries; RAG (retrieval-augmented generation) anchored on URIs	Foundry IQ knowledge bases on Azure AI Search	Amazon Bedrock Knowledge Bases; pgvector (vector search in PostgreSQL)

Agents	Tool calls, delegation, messages	MCP (Model Context Protocol), A2A (Agent2Agent protocol), OpenAPI (standard API description)	Copilot Studio; Microsoft Foundry Agent Service	LangGraph (agent framework); Claude; Workday agents
Telemetry	Every decision and its lineage	PROV-O (W3C Provenance Ontology), OpenTelemetry (open observability standard)	Azure Monitor; Microsoft Sentinel (SIEM: security information and event management); Purview Audit	Datadog; OpenLineage (open data-lineage standard)
Analytics	Governed metrics and data products	DPROD (Data Product Ontology, an OMG specification)	OneLake (Fabric's data lake); Power BI	dbt Semantic Layer; Apache Superset

Three points to make aloud.

- **Microsoft's guardrails stay.** Purview DLP, sensitivity labels and Entra Conditional Access remain the perimeter. The Keel adds controls they cannot express: co-employment allocation (which obligations belong to Insperity and which to the client), jurisdiction-specific rules and per-client separation of knowledge.
- **Four architecture decisions back the table.** Canonical truth lives in open standards. Enforcement sits in the request path, outside any model vendor. Every agent has an identity and a governed profile. One adapter swap is exercised at every phase gate.
- **One open item.** Whether Workday agents support MCP and A2A needs confirming. If they do not, Workday agents reach the Keel through the gateway's APIs (application programming interfaces), and the contract still holds.

The closing arc: slides 13–14

The roadmap delivers something visible in every phase, and the talk ends on four decisions rather than a pitch.

Slide 13, four gated phases over 18 months. The phases follow the IIH roadmap from the source materials. No phase opens until the previous gate passes on the PLM (program lifecycle management) Dashboard.

- **Phase 0, Foundation (months 1–2).** Build isp-core, key all 142 rules with an owner, source, and URI, stand up the identity registry, and admit the first 2,000 assets. North Star: a staff search pilot. Gate: at least 98% of admitted assets pass SHACL validation.
- **Phase 1, Governed Copilot MVP (months 3–6).** MVP means minimum viable product. Deliver the payroll and benefits holons, the rule compiler and the gateway with MTBH and the Simplex gate. North Star: cited answers for employees at 3–5 pilot clients. Gate: P95 response time of 3 seconds or less (95% of answers arrive within 3 seconds) and zero cross-tenant incidents.
- **Phase 2, Domain copilots (months 7–12).** All six domain holons, governed personal models and the Workday HRScale adapter. North Star: the compass navigator inside client and employee portals. Gate: MTBH of at least 96% on every dimension, with 25–40 client companies live.
- **Phase 3, Governed autonomy (months 13–18).** Regulatory change discovery and the 50-state rollout. North Star: Insperity's governed HR tools available inside clients' own AI copilots. Gate: MTBH of at least 97%, sustained.

Slide 14, four decisions to start.

1. Charter the Semantic Governance Council, with authority over rule changes. Without it, author-once cannot work.
2. Confirm the Microsoft footprint: Purview, Fabric capacity and Copilot Studio licensing decide which adapters go first.
3. Set the platform order: HRScale clients on Workday versus clients on Insperity payroll decides the scope of parity testing.
4. Bring the compass mark in: the brand team approves the logo as the entry point for search and navigation.

End on the line from the slide: the mark already points north, and the Keel keeps the company on that heading.

Glossary

Every acronym and specialist term in the deck, in alphabetical order.

Term	Stands for	What it means here
5D context bundle	Five dimensions	The WHO, WHAT, WHEN, WHERE and WHY attached to every request and search
A2A	Agent2Agent protocol	Open protocol for AI agents to delegate tasks to one another
AEDT	Automated employment decision tool	AI used in hiring; bias audits required by laws such as NYC Local Law 144
AI	Artificial intelligence	
AICPA	American Institute of Certified Public Accountants	Sets the standards behind SOC reports
API	Application programming interface	How one system calls another
BPA	Business Performance Advisor	Insperity's client-facing sales and advisory role
CA	California	
CI	Continuous integration	Automated build-and-test pipeline; runs the drift check

CPEO	Certified Professional Employer Organization	IRS certification under Internal Revenue Code section 7705
DLP	Data loss prevention	Microsoft Purview controls that stop sensitive data leaving
dbt	data build tool	Framework for testing and transforming warehouse data
DPROD	Data Product Ontology	OMG specification for describing data products
ESAC	Employer Services Assurance Corporation	Independent accreditation body for PEOs
FAMLI	Family and Medical Leave Insurance	Colorado's paid leave program
Holon / holarchy		A part that is also a whole; a hierarchy built from them
HR / HRBP	Human resources / HR business partner	
IIH	Program roadmap name from the source materials	The four-phase roadmap the gates come from

isp-core, isp-pay, isp-ben, isp-comp, isp-tal, isp-tech	Insperity ontology modules	Core, payroll and tax, benefits, compliance and risk, talent, HR technology
JSON	JavaScript Object Notation	Common data format; JSON Schema defines valid responses
KB	Knowledge base	As in Amazon Bedrock Knowledge Bases
KGCL	Knowledge Graph Change Language	Standard record of each change to the ontology
L1–L4	Levels 1–4	Knowledge, domain, gateway and enterprise levels of the Keel
LKIF	Legal Knowledge Interchange Format	Ontology for obligations, permissions and prohibitions
MCP	Model Context Protocol	Open protocol for AI agents to call tools and data
MTBH	Mean Time Between Hallucinations	The program's AI reliability score across five dimensions
MVP	Minimum viable product	
OIDC	OpenID Connect	Standard sign-in protocol

OMG	Object Management Group	Standards body behind DPROD
OPA / Rego	Open Policy Agent / its policy language	Engine that makes allow or deny decisions
OpenAPI		Standard way to describe an API
ORSD	Ontology Requirements Specification Document	States what the model must be able to answer
OWL / owl:imports	Web Ontology Language	W3C ontology standard; owl:imports lets one ontology include another
P95	95th percentile	95% of responses arrive within the stated time
PEO	Professional employer organization	Co-employment provider, such as Insperity
PEP	Policy enforcement point	Checkpoint that allows, denies, holds or redacts a request
PII	Personally identifiable information	
PLM Dashboard	Program lifecycle dashboard	Gate and Go/No-Go console for the 28 program projects

PROV-O	W3C Provenance Ontology	Standard for recording where data and decisions came from
RAG	Retrieval-augmented generation	Grounding AI answers in retrieved documents
RDF	Resource Description Framework	W3C standard for linked statements; the Keel's native form
SCIM	System for Cross-domain Identity Management	Standard for syncing user accounts
SHACL	Shapes Constraint Language	W3C standard for validating data against rules
SIEM	Security information and event management	Category of Microsoft Sentinel
Simplex gate		Human approval hold on irreversible actions
SKOS	Simple Knowledge Organization System	W3C standard for vocabularies and taxonomies
SOC 1 / SOC 2	System and Organization Controls	Independent audits of financial-reporting and security controls
SPARQL	SPARQL Protocol and RDF Query Language	Query language for knowledge graphs

SQL	Structured Query Language	Standard database query language
SSOT	Single source of truth	The 142-rule policy register
SWRL	Semantic Web Rule Language	If-then rules over an ontology
URI	Uniform Resource Identifier	Permanent unique name for a worker, client or rule
W3C	World Wide Web Consortium	Web standards body